

CIVIL ENGINEERING SERIES ANALYSIS & DESIGN REPORT

Author: Structural Engineering Department • **Reference:** Infinite Series Approximation Study

1. Executive Summary & Problem Scope

This report evaluates the application of infinite series expansions (Geometric, Power, Maclaurin, and Taylor series) in modeling structural trigonometric quantities ($y = L \sin \theta$). By implementing explicit iterative loops in Python without relying on built-in trigonometric shortcuts, this study analyzes truncation error behavior, convergence rates, and optimal term selections for civil engineering design.

2. Geometric & Power Series Convergence Dynamics

- Partial Sum Behavior:** Summing partial terms $S_N = \sum x^k$ iteratively demonstrates how higher-order polynomials capture progressive magnitude contributions.
- Convergence Rate Variance:** Inputs near zero ($x = 0.5$) exhibit rapid exponential decay, achieving tight convergence within 4 terms. Conversely, inputs approaching unity ($x = 0.9$) require extended term depths because higher-order exponents decay at diminishing rates.

3. Maclaurin Series & Structural Angle Investigation

- Error Progression Across Angles:** Evaluating $y = 20 \sin \theta$ shows that Maclaurin polynomials centered at 0° maintain exceptional precision for minor inclinations ($\leq 5^\circ$), but truncation discrepancies expand non-linearly as angles exceed 20° .
- Term Depth Optimization:** For small angles ($< 5^\circ$), a 1-term linear approximation ($\sin \theta \approx \theta$) satisfies standard $< 0.1\%$ error thresholds. For broader spans up to 30° , at least 3 terms are necessary to suppress higher-order remainder terms.

Table 1: Numerical Verification for Selected Angles

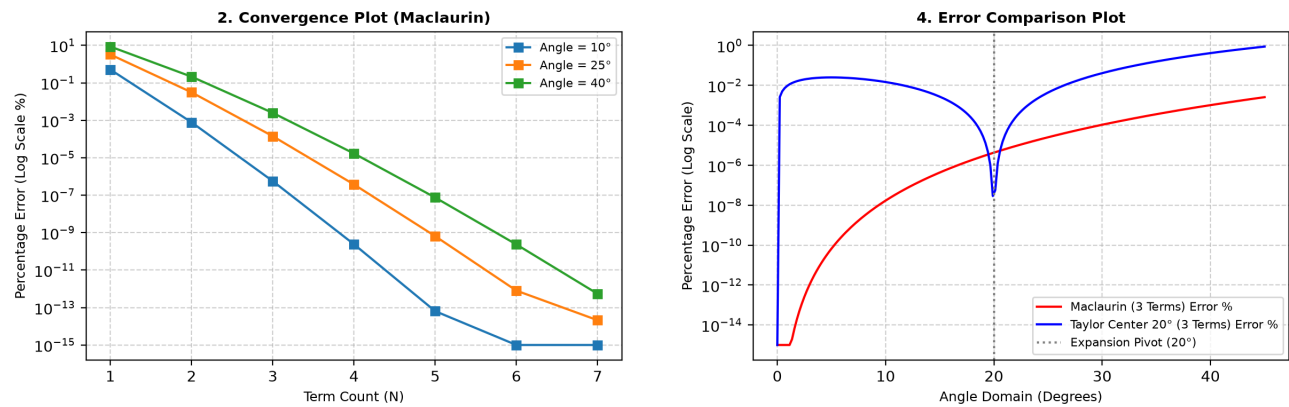
Angle	Terms (N)	Exact (m)	Approx (m)	Error (%)
10°	2	3.47296	3.47294	0.0008%
10°	4	3.47296	3.47296	0.0000%
10°	6	3.47296	3.47296	0.0000%
25°	2	8.45237	8.44974	0.0310%
25°	4	8.45237	8.45237	0.0000%
25°	6	8.45237	8.45237	0.0000%
40°	2	12.85575	12.82843	0.2125%
40°	4	12.85575	12.85575	0.0000%
40°	6	12.85575	12.85575	0.0000%

4. Taylor Series Expansion & Error Tolerance Evaluation

- Shifted Pivot Advantage:** Centering a Taylor series at $a = 20^\circ$ redistributes polynomial accuracy, drastically minimizing remainder errors around the structural pivot while sacrificing precision near the origin (0°).

- Critical Small-Angle Boundary:** The single-term small-angle approximation ($\sin \theta \approx \theta$) violates the 0.1% error tolerance threshold at approximately 4.45° , beyond which cubic curvature terms become mandatory.

Figure Visualizations (Convergence & Error Profiles):



5. Final Engineering Design Recommendation

To achieve robust computational performance with $<0.1\%$ truncation error across general structural surveying modules:

- Algorithmic Selection:** For standard localized subroutines restricted to angles under 25° , a 3-term Maclaurin expansion provides the ideal balance of arithmetic simplicity and precision.
- Shifted Regional Strategy:** For wide-span or inclined structural subsystems offset from the origin, engineers should deploy localized Taylor expansions anchored to regional pivot nodes.
- Conclusion:** Explicit loop-based polynomial evaluations offer complete transparency and eliminate software black-box dependencies, making them highly reliable for custom structural analysis toolkits.